1. The task is to apply and analyze the watermarking algorithm (https://arxiv.org/pdf/2301.10226.pdf) on a pre-trained summarization model (T5, BART, Flan, or others) and a summarization dataset (CNN, Xsum, or others).

Analyzing Watermark Algorithm (lm-watermarking) on a pre-trained summarization model (T5) with summarization datasets (CNN/Daily Mail, XSum)

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Execution Code Link:
https://colab.research.google.com/drive/1L3a-VJGJEkN-NWBqHTTo2iLIOYme5NRq?usp=sharing

A Watermark for Large Language Models [1], is a framework that mitigate potential harms associated with large language models (LLMs) by embedding signals into the generated text that are invisible to humans but algorithmically detectable.

The implementation of watermarking in language models involves two main algorithms: one for embedding the watermark during text generation and another for detecting the watermark in the generated text. In this analysis process, I analyze the watermarking algorithm for detecting the watermark on a pre-trained summarization model, i.e., Google Text-to-Text Transfer Transformer (T5 small) with summarization datasets (CNN/Daily Mail, XSum).

In this study, first I imported the datasets and the necessary libraries. Then created a method called `get_wm_pred(text)` that sends the texts to `WatermarkDetector(...)` that detects the watermark for generated texts and classifies it as Human/Non-Watermarked or Watermarked as described in the source paper [1]. Through this method, the data from CNN/Daily Mail (['article', 'highlights', 'id']) dataset can be passed to predict the labels, i.e., human written texts labeled as "0" or LLM generated summarized texts labeled as "1".

Next step, I load the LLM model - Text-to-Text Transfer Transformer (T5) [2] for text summarization. Here, the model generates summarized texts from the datasets (CNN/Daily Mail, XSum) with the given logit processor method - `WatermarkLogitsProcessor(...)`. Now, these summarized texts are sent to `WatermarkDetector(...)` method to detect whether the texts are Human written texts labled as "0" or LLM generated texts labeled as "1".

Then, I evaluated the results through Confusion Matrix (TP, FP, TN, FN).

Table I
Confusion Matrix

		Predicted	
		Positive (P)	Negative (N)
Ground Truth	True (T)	True Positive (TP)	False Negative (FN)
	False (F)	False Positive (FP)	True Negative (TN)

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \qquad (1)$$

$$\text{Recall (TPR)} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (2)$$

$$\text{Precision} = \frac{\text{TP}}{TP + \text{FP}} \quad (3)$$

$$\text{F1 score} = 2 \times \frac{\text{recall} \times \text{precision}}{\text{recall} + \text{precision}} \quad (4)$$

$$\text{False Positive Rate (FPR)} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (5)$$

Experiment #1: In the first run, I considered 200 entries from CNN/Daily Mail datasets. With this experiment, all 200 out of 200 human generated texts were classified correctly but only 62 out of 200 LLM generated texts were classified correctly. Figure 1 shows the confusion matrix of experiment with CNN/Daily Mail datasets and Table II describes the resulted classification metrics.

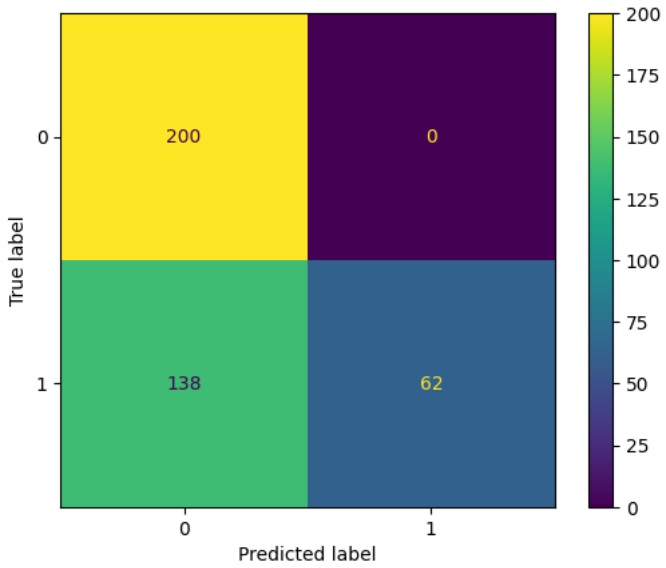


Figure 1. Resulted confusion matrix for watermark algorithm analysis with human generated texts and LLM generated texts prediction with CNN/Daily Mail datasets.

Table II
Classification Report for CNN/Daily Mail datasets with Model T5 generated summarized texts

	Classification Report			
	precision	recall	f1-score	accuracy
Human Generated Texts (labeled as 0)	0.59	1.00	0.74	0.66
LLM Generated Summarized Texts (labeled as 1)	1.00	0.31	0.47	

This classification report shows that the model achieves an overall accuracy of 66%, indicating moderate performance in distinguishing human-written texts and LLM generated summarized texts. The model exhibits a strong performance for classifying human generated written texts, achieving a high recall of 100% for human text but at the cost of low precision (59%), meaning it often misclassifies LLM-generated text as human. Conversely, while its precision for LLM generated summarized texts (100%) is perfect, it fails to detect a majority of LLM-generated samples, leading to a low recall (31%). This imbalance suggests the model is overly cautious in predicting LLM-generated text and may benefit from parameter calibration to improve its ability to distinguish between the two classes effectively.

Experiment #2: In this run 200 entries are considered from XSum datasets. With this experiemnt, again all human generated texts are classified correct (200/200) but the detector failed to classify LLM generated texts correctly (49/200) less than CNN/Daily Mail dataset. Figure 2 shows the confusion matrix of experiment with XSum datasets and Table III describes the resulted classification metrics.

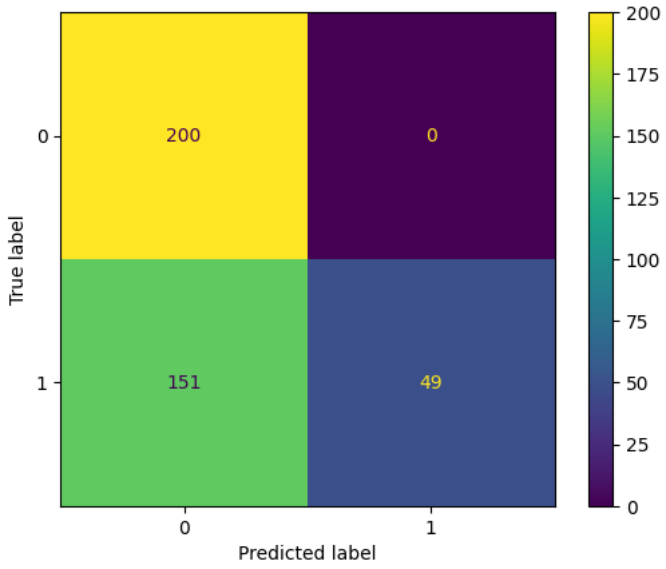


Figure 2. Resulted confusion matrix for watermark algorithm analysis with human generated texts and LLM generated texts prediction with XSum datasets.

Table III
Classification Report for XSum datasets with Model T5 generated summarized texts

	Classification Report			
	precision	recall	f1-score	accuracy
Human Generated Texts (labeled as 0)	0.57	1.00	0.73	0.62
LLM Generated Summarized Texts (labeled as 1)	1.00	0.24	0.39	

Finally, I compare and plotted accuracy of two experiments with different datasets for watermark algorithm detection. Figure 3 depicts the comparison result.

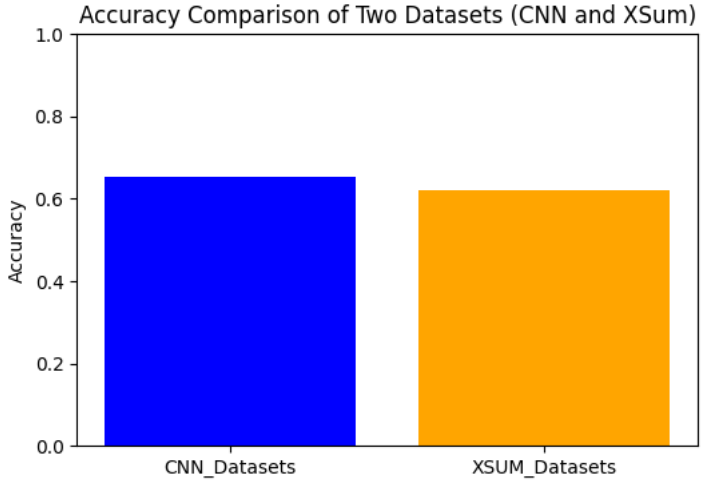


Figure 3. Comparison results of accuracy for two different datasets (200 samples) with the LLM model Google T5.

Overall Observations:

- The model is highly confident when predicting LLM generated but fails to detect most of them (low recall indicates that).
- The model is strongly biased toward predicting human generated texts, as seen in the high recall for the human class but extremely low recall for the LLM generated summarized watermarked text class.
- The overall accuracy of 66% on the CNN/Daily Mail dataset and 62% on the XSum dataset suggests that the model is better than random guessing but is not performing optimally in distinguishing between human-written and LLM-generated texts on these datasets.

Reference:

[1] J. Kirchenbauer, J. Geiping, Y. Wen, J. Katz, I. Miers, and T. Goldstein, "A watermark for large language models," in *International Conference on Machine Learning*, 2023: PMLR, pp. 17061-17084.

[2] C. Raffel *et al.*, "Exploring the limits of transfer learning with a unified text-to-text transformer," *Journal of machine learning research,* vol. 21, no. 140, pp. 1-67, 2020.